

Sweet and sour cherry phenolics and their protective effects on neuronal cells.



The identification of phenolics from various cultivars of fresh sweet and sour cherries and their protective effects on neuronal cells were comparatively evaluated in this study. Phenolics in cherries of four sweet and four sour cultivars were extracted and analyzed for total phenolics, total anthocyanins, and their antineurodegenerative activities. Total phenolics in sweet and sour cherries per 100 g ranged from 92.1 to 146.8 and from 146.1 to 312.4 mg gallic acid equivalents, respectively. Total anthocyanins of sweet and sour cherries ranged from 30.2 to 76.6 and from 49.1 to 109.2 mg cyanidin 3-glucoside equivalents, respectively. High-performance liquid chromatography (HPLC) analysis revealed that anthocyanins such as cyanidin and peonidin derivatives were prevalent phenolics. Hydroxycinnamic acids consisted of neochlorogenic acid, chlorogenic acid, and p-coumaric acid derivatives. Glycosides of quercetin, kaempferol, and isorhamnetin were also found. Generally, sour cherries had higher concentrations of total phenolics than sweet cherries, due to a higher concentration of anthocyanins and hydroxycinnamic acids. A positive linear correlation ($r^2 = 0.985$) was revealed between the total anthocyanins measured by summation of individual peaks from HPLC analysis and the total anthocyanins measured by the pH differential method, indicating that there was in a close agreement with two quantifying methods for measuring anthocyanin contents. Cherry phenolics protected neuronal cells (PC 12) from cell-damaging oxidative stress in a dose-dependent manner mainly due to anthocyanins. Overall results showed that cherries are rich in phenolics, especially in anthocyanins, with a strong antineurodegenerative activity and that they can serve as a good source of biofunctional phytochemicals in our diet.